

MTP-1 PRESENTATION
ON

Virtual Fabrication and Investigation of Electrical Characteristics of Si/SiGe Heterostructure Based Bulk FinFET

*Presented By :
Rishav Kumar
2411EE37
VLSI & Embedded System*

List of Contents

- Moore's Law
- Why FinFET?
- Introduction
- Motivation and Objective
- Proposed Structure
- Why Si/SiGe
- Advantages and Uses of Si/SiGe
- Process flow of virtual fabrication
- Result Discussion
- Conclusion
- Future Work
- References

Moore's Law

- Number of transistors on an Integrated Circuit (IC) doubles about every two year.
- This was possible due to advent of metal-oxide-semiconductor(MOS) Technology.

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count

50,000,000,000

10,000,000,000

5,000,000,000

1,000,000,000

500,000,000

100,000,000

50,000,000

10,000,000

5,000,000

1,000,000

500,000

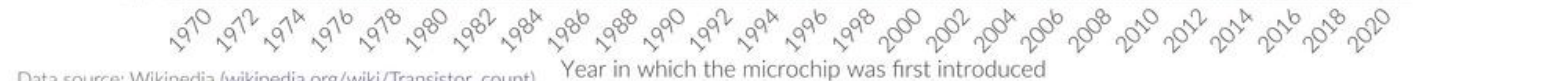
100,000

50,000

10,000

5,000

1,000



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

A semi-log plot of transistor counts for microprocessors against dates of introduction, nearly doubling every two years

Why FinFET?

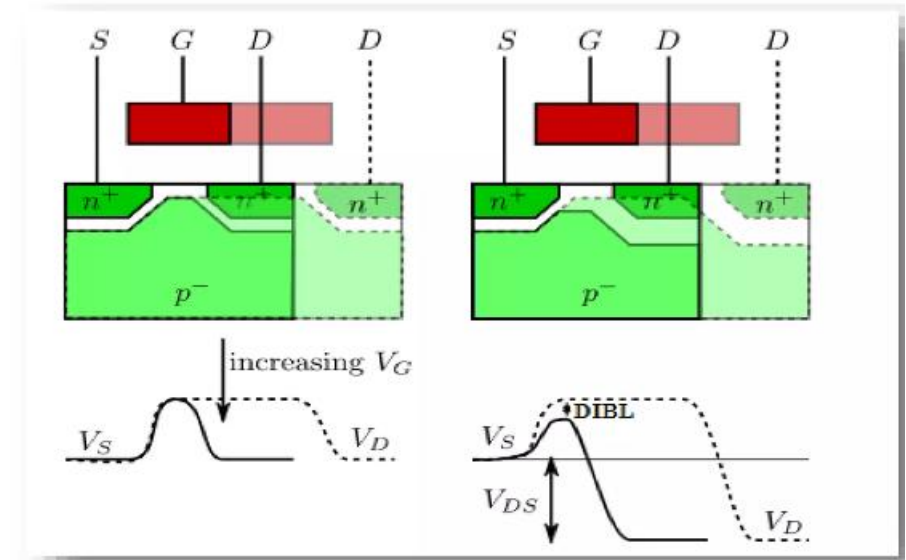
- Because of continuous scaling down of channel length of MOSFET, many **short channel effects** dominate.
- **Short Channel Effects** : When the channel length of MOSFET becomes the same order of magnitude as the **depletion layer width of source and drain**, the transistor starts to behave differently leading to performance impact.
- Drain Induced Barrier Lowering(DIBL), velocity saturation, impact ionization, hot electron effect are some of the examples of short channel effects.

Drain Induced Barrier Lowering (DIBL) :

- Drain Induced Barrier Lowering (DIBL) refers to the phenomenon in MOSFET, where the Threshold Voltage (V_{th}) reduces because of the increase in applied Drain Voltage (V_{DS}).
- Ideally, only Gate Voltage (V_{GS}) should be responsible for lowering the barrier to the point where electron is able to flow from source to drain. However, as the channel length shortens, the applied potential on drain increases depletion region to a point that reduces potential barrier thus lowering threshold voltage.

Thus, in short channel : Channel length ↓
 $V_{DS} \uparrow \longrightarrow$ Effective Channel Length ↓
Because of this, **the barrier lowers**, and thus $V_{TH} \downarrow$

- Because of DIBL, device can turn on even for Gate Voltage less than Threshold Voltage.



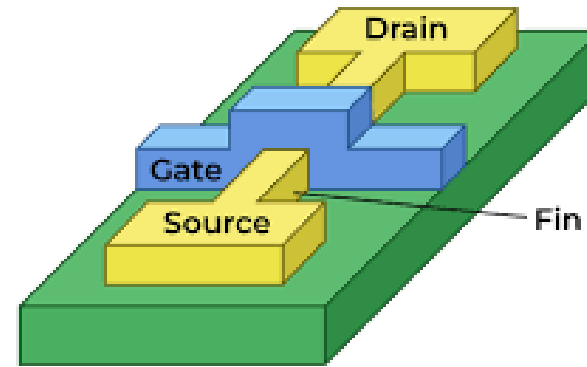
---- : Long Channel Device
— : Short Channel Device

Left : Low V_{DS}
Right : High V_{DS}

- So, because of DIBL and other short channel effects, the performance of MOSFET ,once the channel starts to reduce, degrades significantly.
- One of the major regions of SCEs was that reduction of control of Gate over channel once channel length reduces.
- Thus, in order to solve this problem, **multi-gate device** was innovated.
- Two or more gates around the channel, provides more electrostatic control over channel, thus supressing SCEs and improving device performance.
- CMOS is the planar technology, where source, drain and channel lies on the same plane.
- Multi-gate technology also led to development of non-planar or “3D” devices where channel lies on different plane than that of source and drain.
- In Double-Gate(DG) MOS technology, gate is two present on two sides, in tri-gate technology, gate is present on three sides of channel.
- Fin-FET technology is tri-gate technology.

Introduction

FinFET (Fin Field Effect Transistor)



- A **3D structure** of MOSFET where the channel is shaped like a **fin** for better control of current.
- Reduces **leakage current** and improves **switching speed** and **power efficiency** compared to traditional planar MOSFETs.
- Widely used in **advanced nodes (<20 nm)** in modern **CPUs and GPUs** for higher performance and lower power consumption.

Motivation and Objective

Motivation :

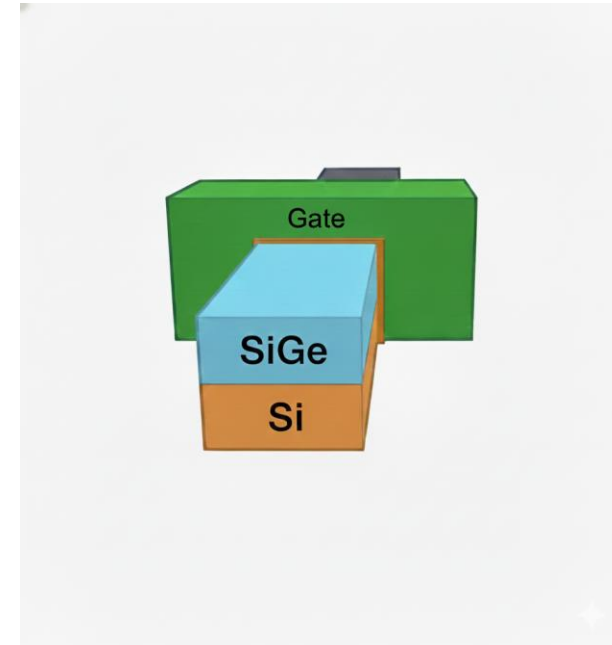
- Modelling and Simulation of FinFET arises from the fact that it offers better channel control, smaller size and improved scalability which made it so popular in industrial applications.

Objective :

- To model heterostructure Si-SiGe channel FinFET technology and simulate it to verify its electrical properties and compare the output with conventional single Si channel FinFET using sprocess and sdevice tool (Synopsys TCAD).

Proposed Structure

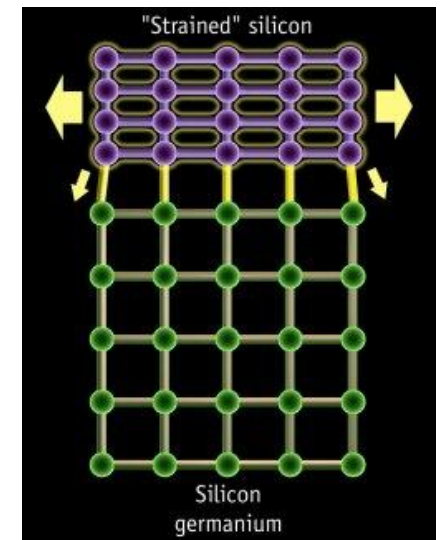
- Conventionally, single material Si channel is in use.
- Proposed structure uses heterojunction material.
- Instead of only Si, SiGe-Si is used to fabricate channel.



**Proposed structure of heterojunction
Si/SiGe channel FinFET**

Why Si-Ge?

- Lattice Constant of pure Si : 5.43 Å
- Lattice Constant of SiGe varies from : 5.43 Å (100% Si) to 5.66Å (100% Ge).
- Introduction of SiGe in Si lattice, leads to change in spacing between atoms.
- This disparity in lattice constant leads to increase in **compressive stress** on the channel.
- Finally, **hole mobility increases**, thus increasing device current.



Strained Si-Ge

Advantage and Uses:

Advantages of SiGe:

- Higher hole mobility
- Enhanced I_{ON} current
- Low temperature compatibility.
- Good ESD and reliability performance

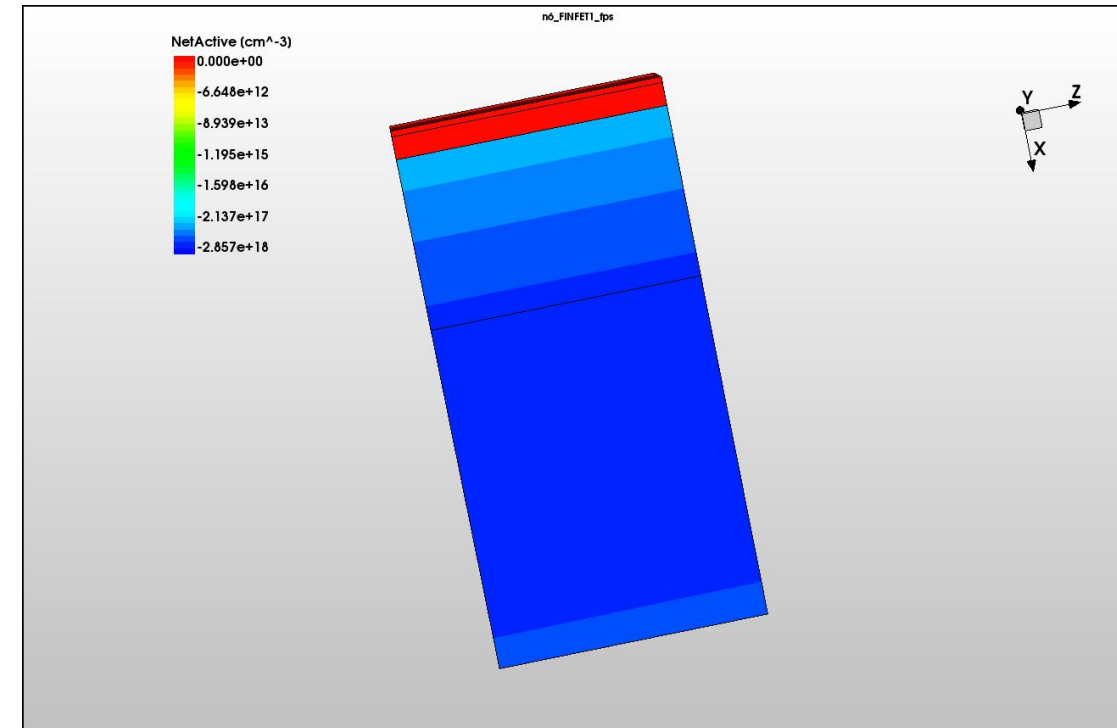
Uses of SiGe heterostructure device :

- High Voltage I/O devices.
- Analog and RF front ends.
- ESD protection structures.
- Mixed Signal Circuits.

Process Flow of Virtual Fabrication

Step 1. P-well implementation

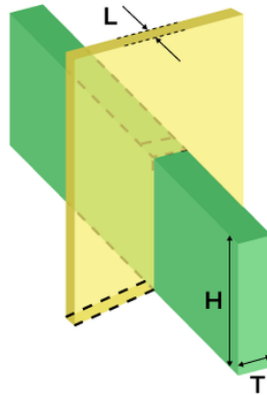
- p-type bulk wafer is deposited.
- 8 nm Silicon and 2nm Silicon-Germanium with mole-fraction of 0.3 is deposited above wafer.
- Doped using ion-implementation with an energy of 30 keV.
- Thermal annealing (RTA) at 1000°C is performed to avoid damage of high energy ions used during ion-implementation.



Step 2. Fin Fabrication using SIT

Because of high aspect ratio of FinFET, it is difficult to realise fins using **photolithography** technique.

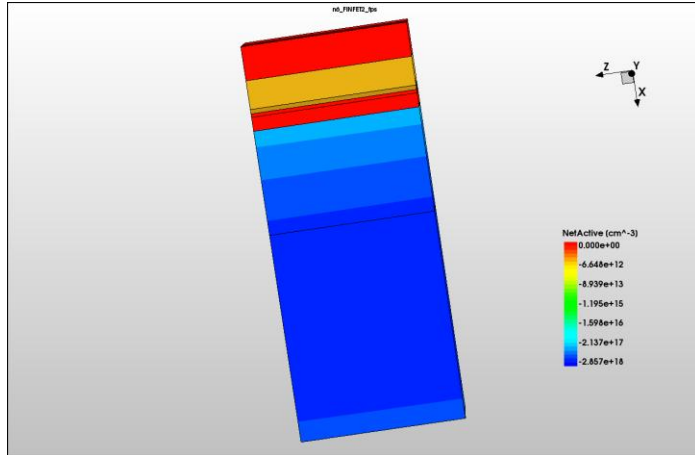
Aspect ratio of a FinFET is defined as the ratio of FH/FT , where FH is Fin Height and FT is Fin Thickness.



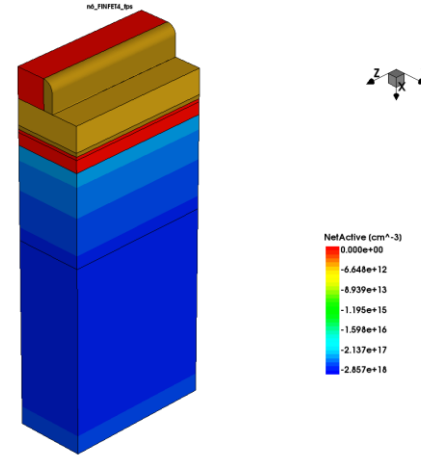
SIT (Sidewall Image Transfer) technique is used to create very narrow and uniform fin with high precision mainly at **nanometre-scale** node.

It consists of 6 steps mainly of deposition and etching of Oxide, Nitride and Silicon to create fin.

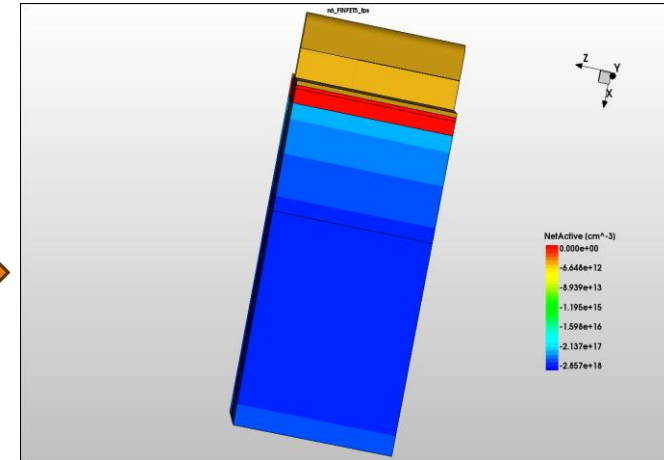
Contd...



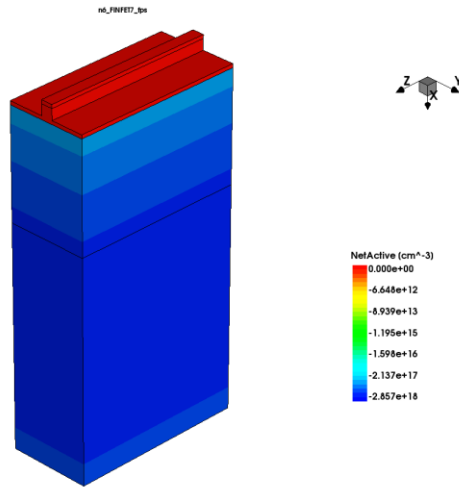
Si sub is covered with SiO₂(2.5nm)(Buffer Oxide),Nitride(16.5nm)(Transfer Layer), Silicon(19.5nm)(Mandrel Layer) by deposition.



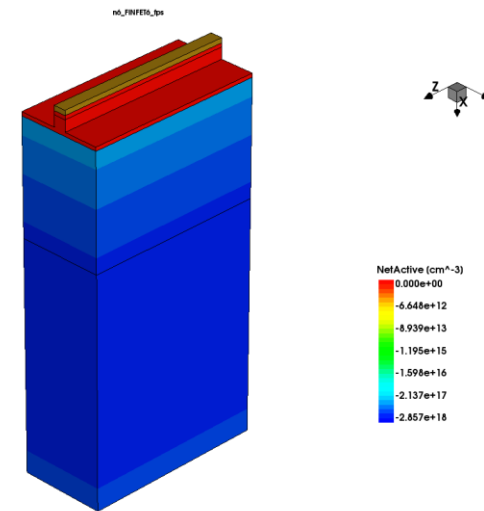
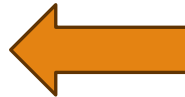
Mandrel is then etched using masking. SiO₂ spacers are formed using anisotropic etching and isotropic overetch.



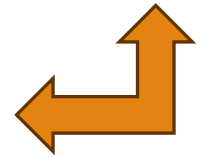
Mandrel is etched. Using spacers as hardmask, transfer layer is also etched.



Oxide is also etched out. Fin Formation is complete.

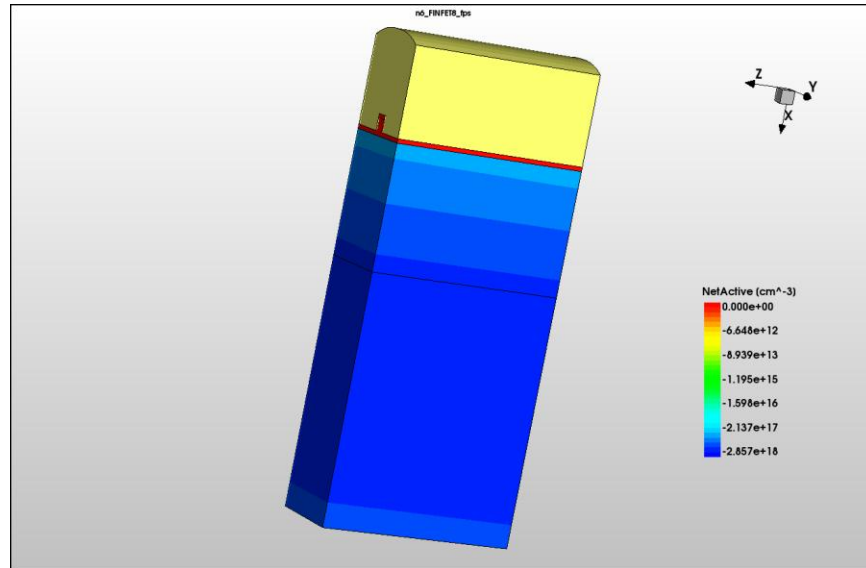


Spacer and Nitride is also etched.

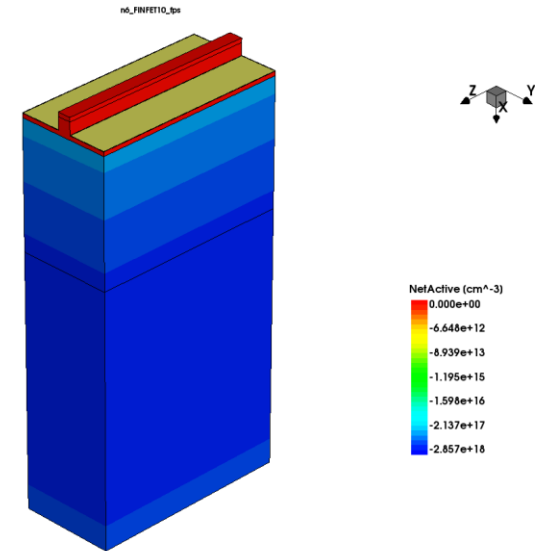


Step 3. TEOS CMP

- Deposition and Etching of Tetraethyl Orthosilicate ($\text{Si}(\text{C}_2\text{H}_5\text{O})_4$).
- TEOS is deposited in CVD(Chemical Vapour Deposition) system and then heated such that it distributes evenly parallel to wafer.
- TEOS surface is smoothed by CMP(Chemical Mechanical Polishing)



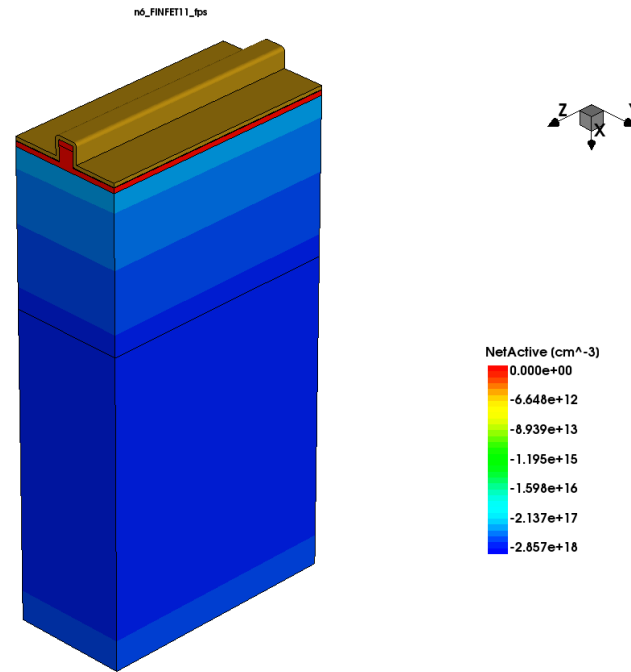
TEOS Deposition



TEOS CMP

Step 4. Thermal Oxidation

- Fin is oxidised using Thermal Oxidation.
- Thermal Oxidation : Oxide grown on the surface using Silicon present at 900°C – 1100°C.

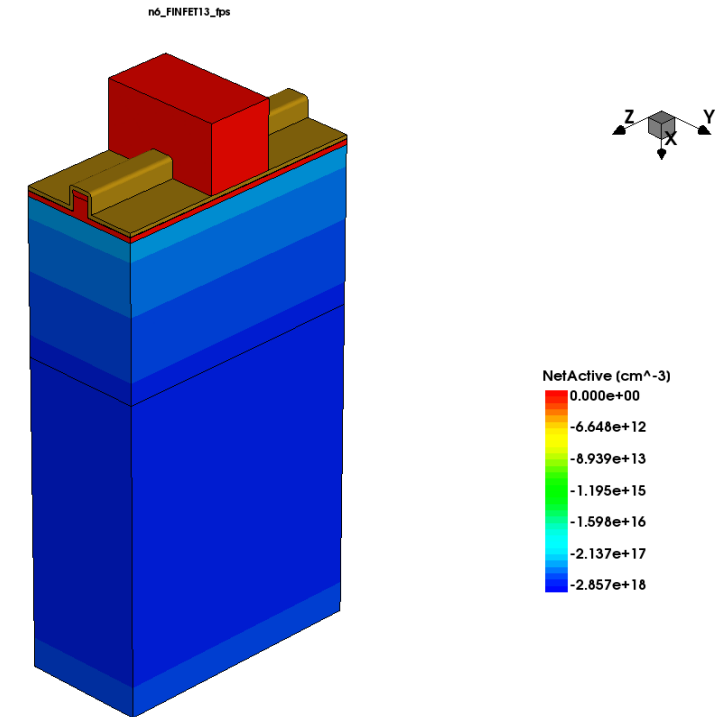


Thin oxide layer is grown using Thermal Oxidation.

Step 5. Dummy Gate Realization

- It is used to favour the realization of Source and Drain extensions and growth.

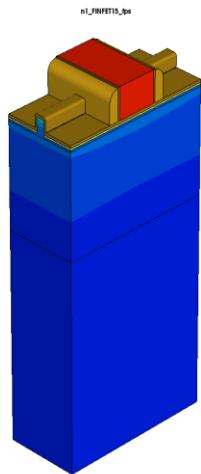
- Layer of Poly-Si is deposited.
- And then etched out such that central part of Fin is covered with Poly-Si.
- Exposed part of Fin is where Source and Drain growth will be realised epitaxially.



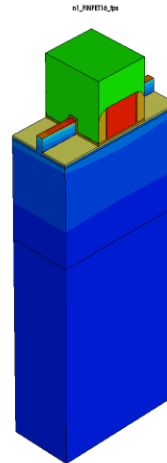
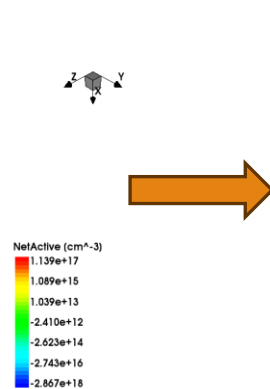
Poly-Si deposited and then etched out to support S/D growth.

Step 6. Source/Drain Creation

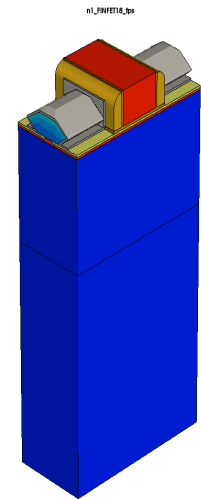
- Nitride Spacers are formed first using SIT process(deposition & etching).
- Then, oxide is removed using Silicon Carbonitride (SiCN) as mask.
- Finally, after removal of oxide, epitaxial growth of Source and Drain is performed.



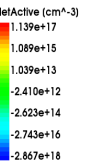
Formation of Nitride Spacer at the sides of dummy gate poly-Si



Removal of oxide on exposed part of fin using SiCN as mask.



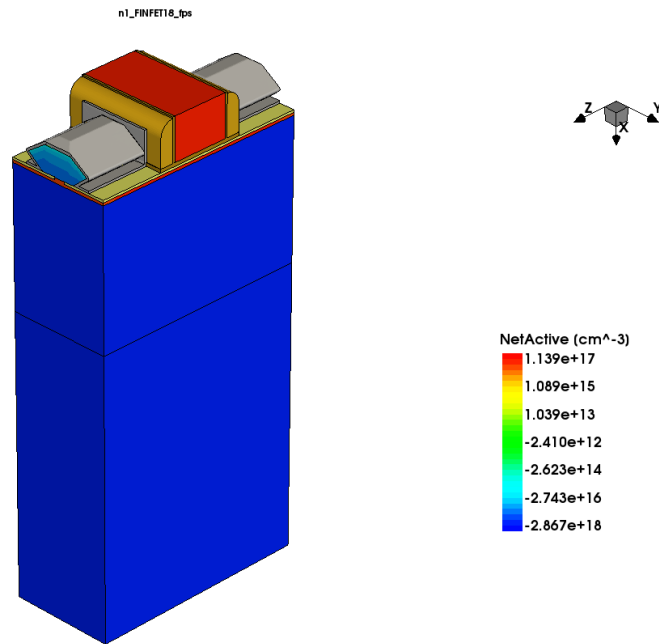
Epitaxial Growth of Source and Drain.



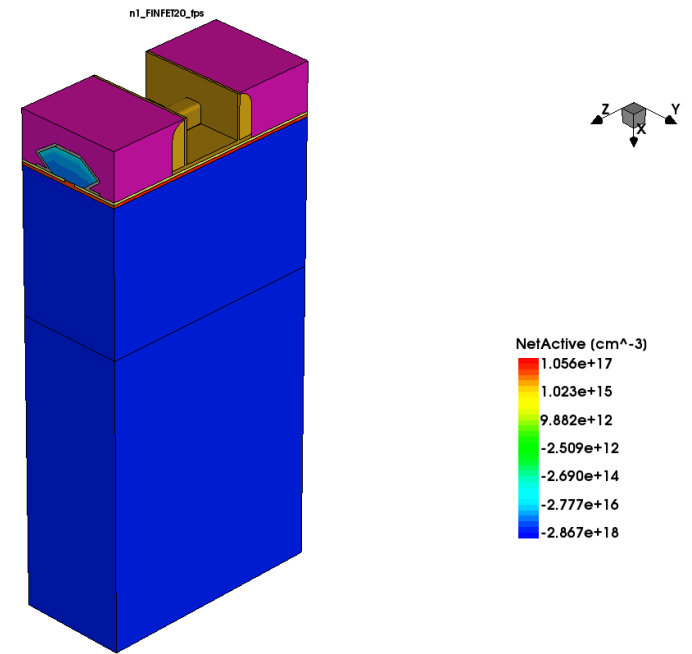
Step 7.Silicidation and Dummy Gate Removal

- SiCN is removed.
- Thin 1 nm layer of Ti is deposited on S/D.
- Ti reacts with Si to give Titanium Silicide.
- Unreacted Titanium is etched.
- Done to reduce Source Drain **Contact resistance**.
- Phosphosilicate Glass(PSG) (20 nm) is deposited all over except dummy gate.
- Dummy Gate is eliminated using Silicon Dioxide as mask.

Contd...



Silicidation



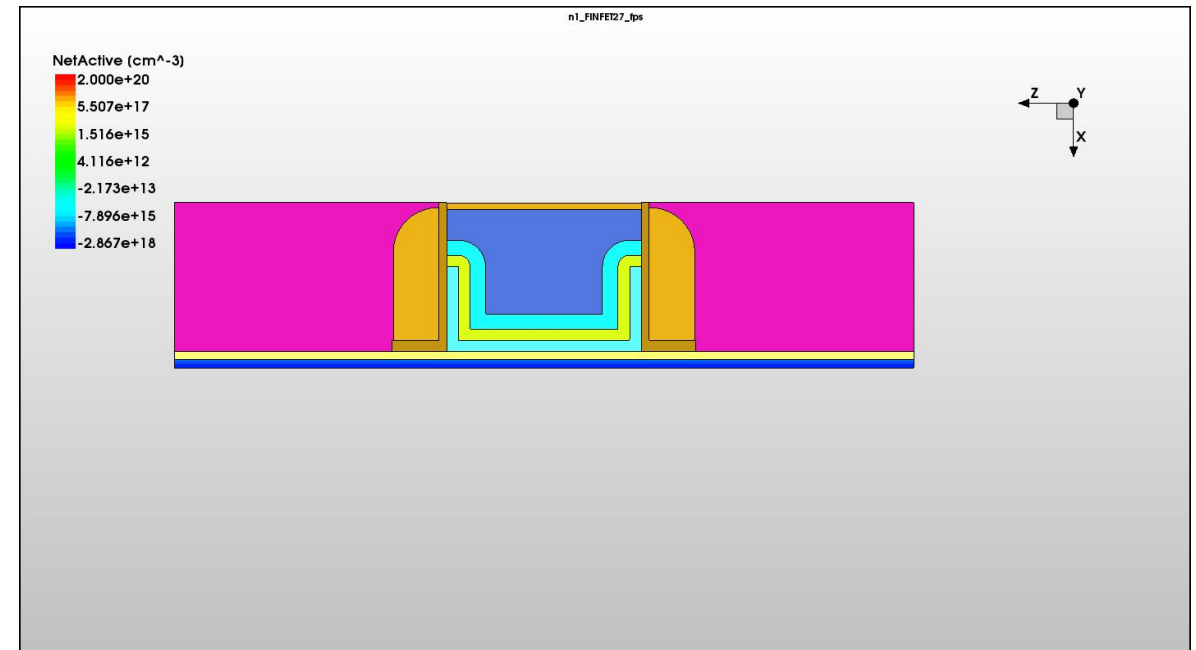
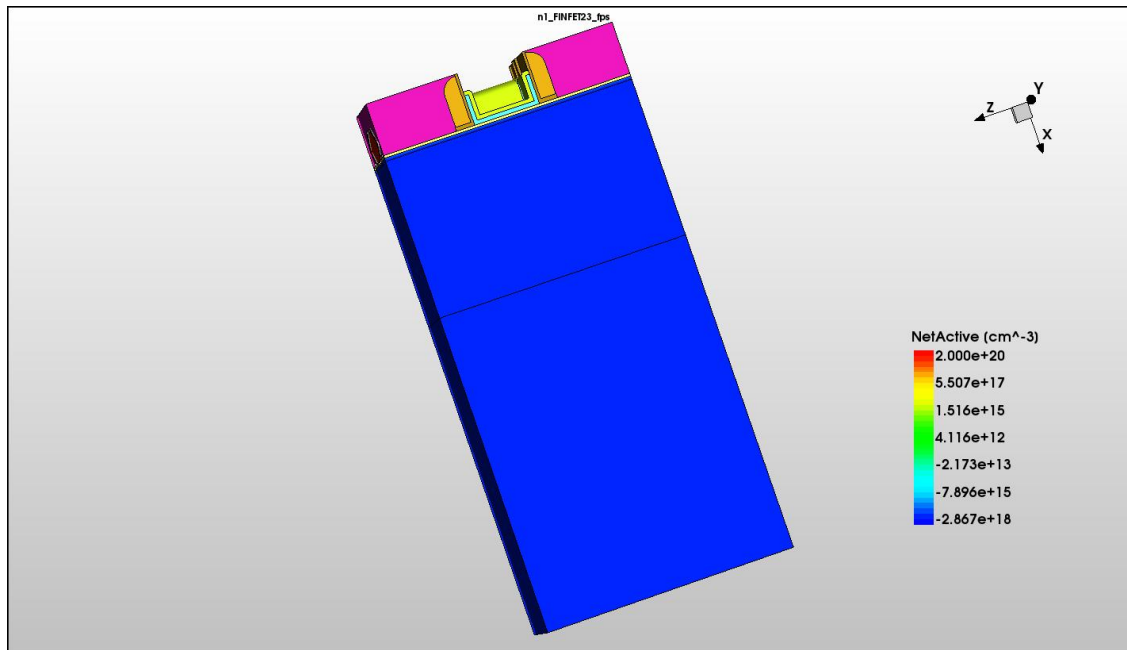
Removal of Dummy Gate

Step 8. Gate Stacking

It is done to avoid dielectric breakdown and to prevent parasitic phenomena such as gate poly depletion.

Stacking is done using Metal-interdiffusion Technology.

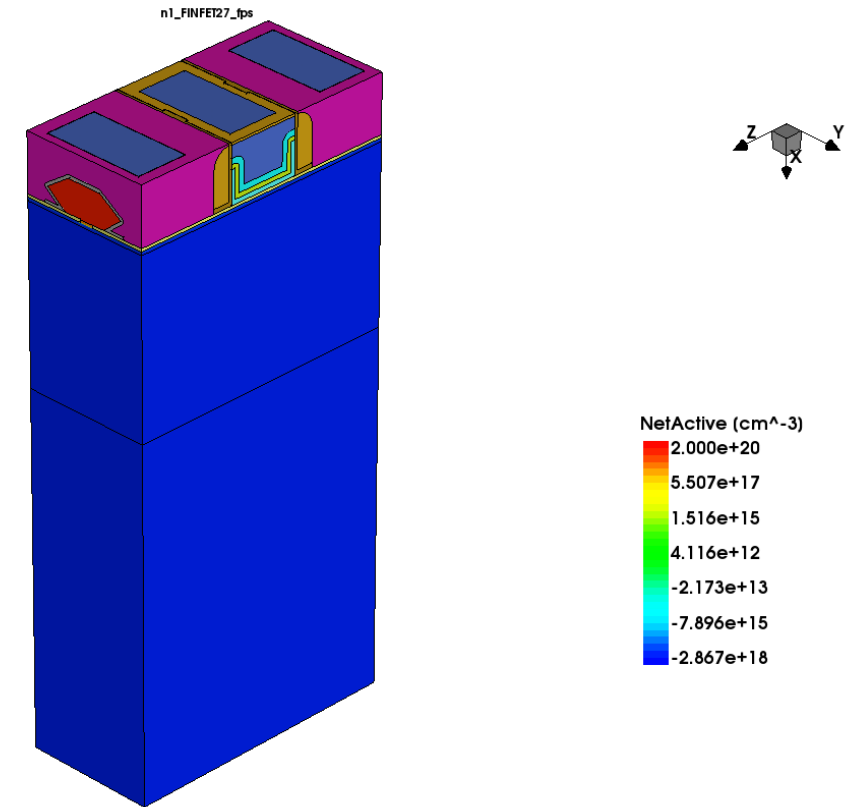
Stack consist of two layers of insulating materials(SiO_2 and HfO_2) and three layers of conductor materials(TiN , TaN , TiAl).



Step 9. Contact

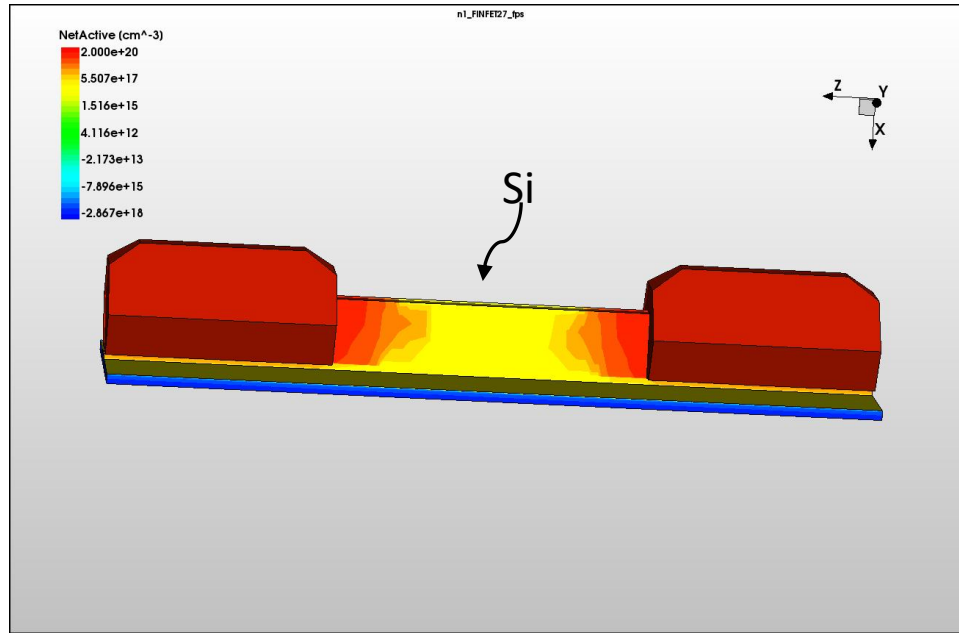
At last, contacts are created with the help of Tungsten deposition and subsequent etching.

Finally, the complete structure of Si/SiGe heterostructure based FinFET is complete.



FinFET complete structure

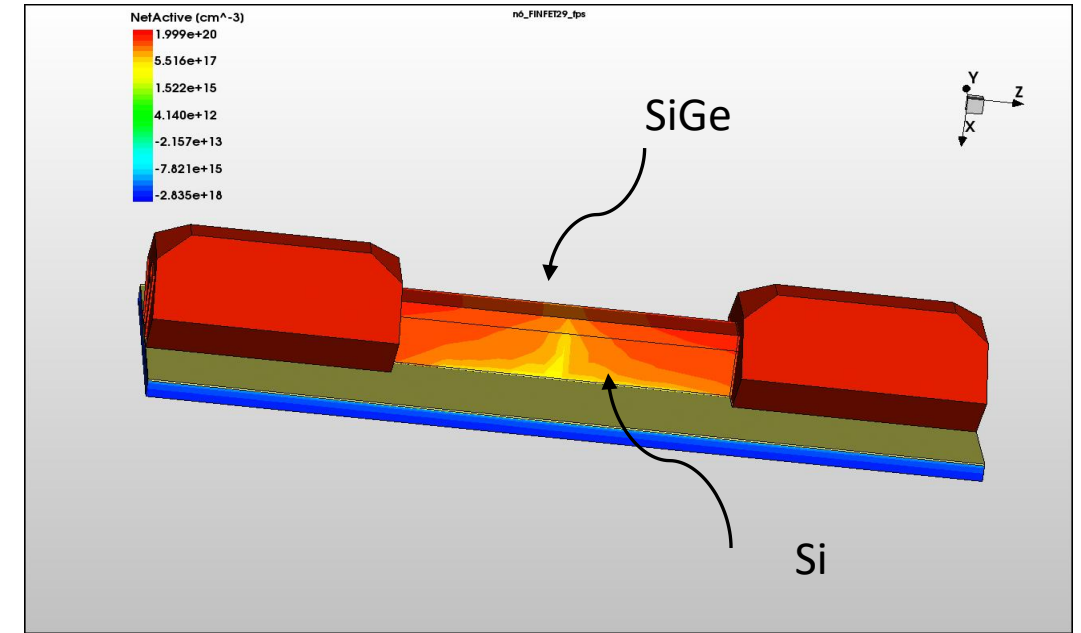
Fin Structure



Fin of the FinFET when fabricated using single material channel of Silicon only.

Fin Height = 10nm.

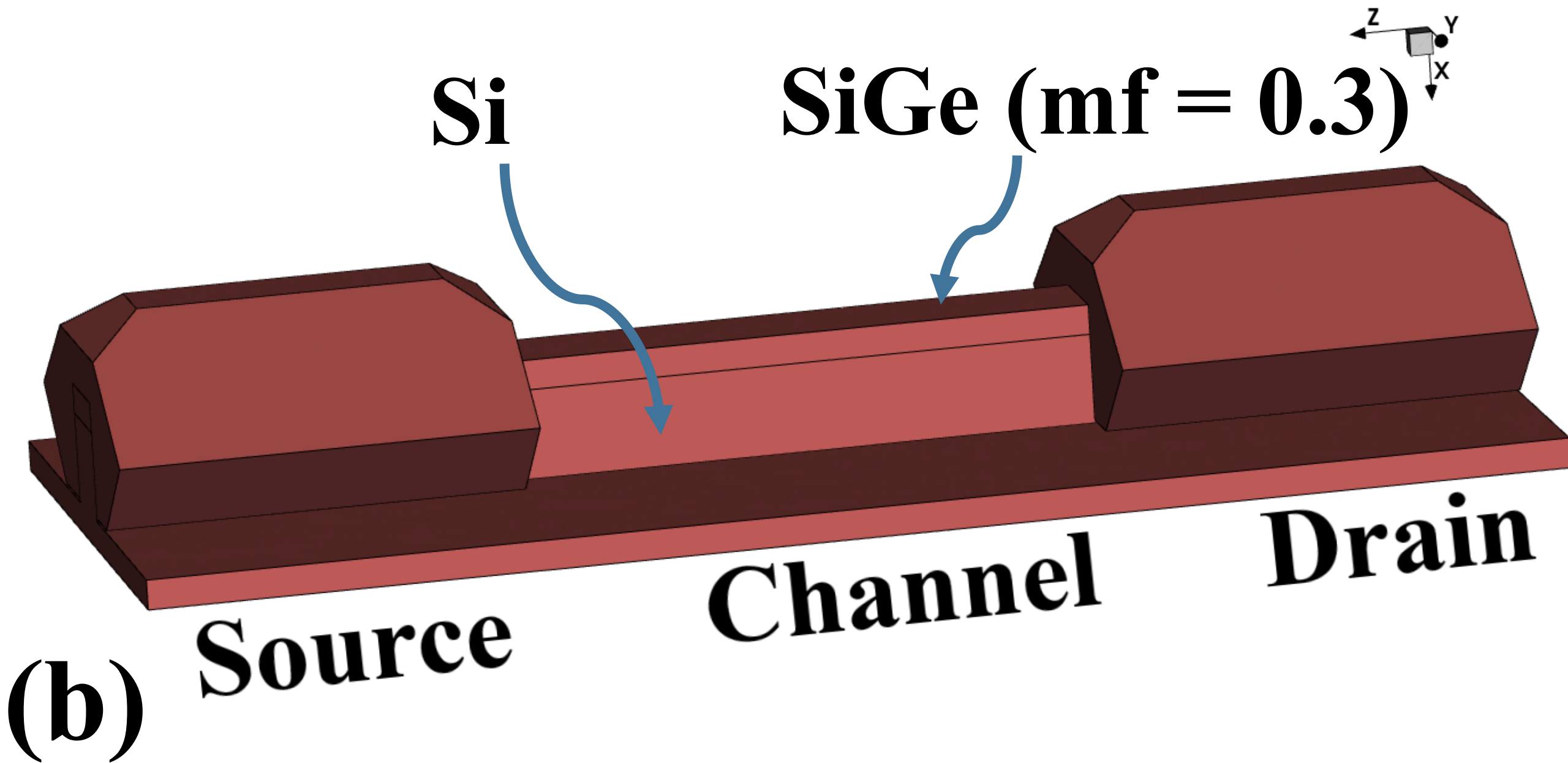
Channel Length = 25nm.

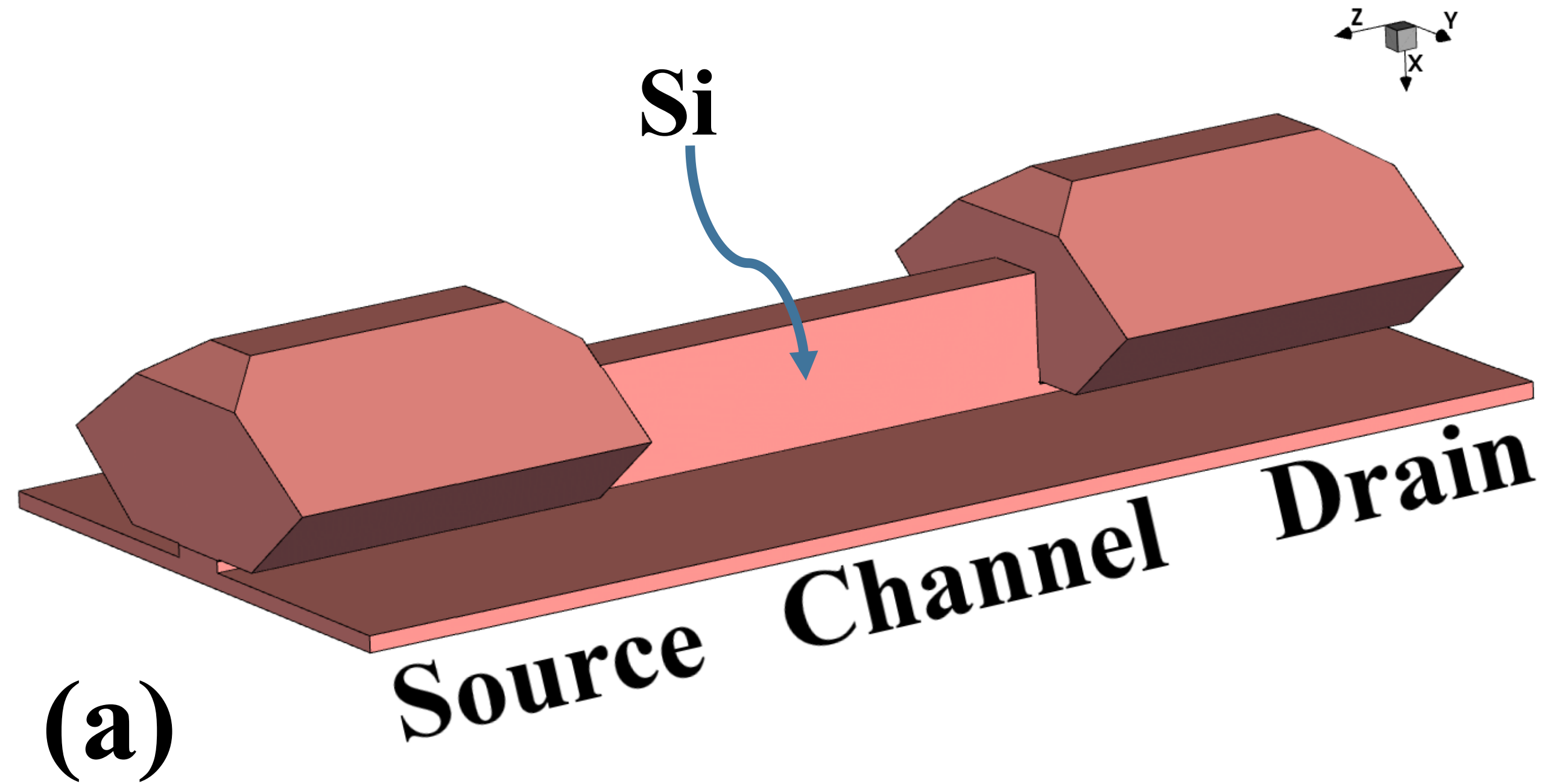


Fin of the FinFET when Fabricated using heterojunction channel of Si and SiGe.

Fin Height = 8 nm Si and 2 nm SiGe (Mole-Fraction = 0.3).

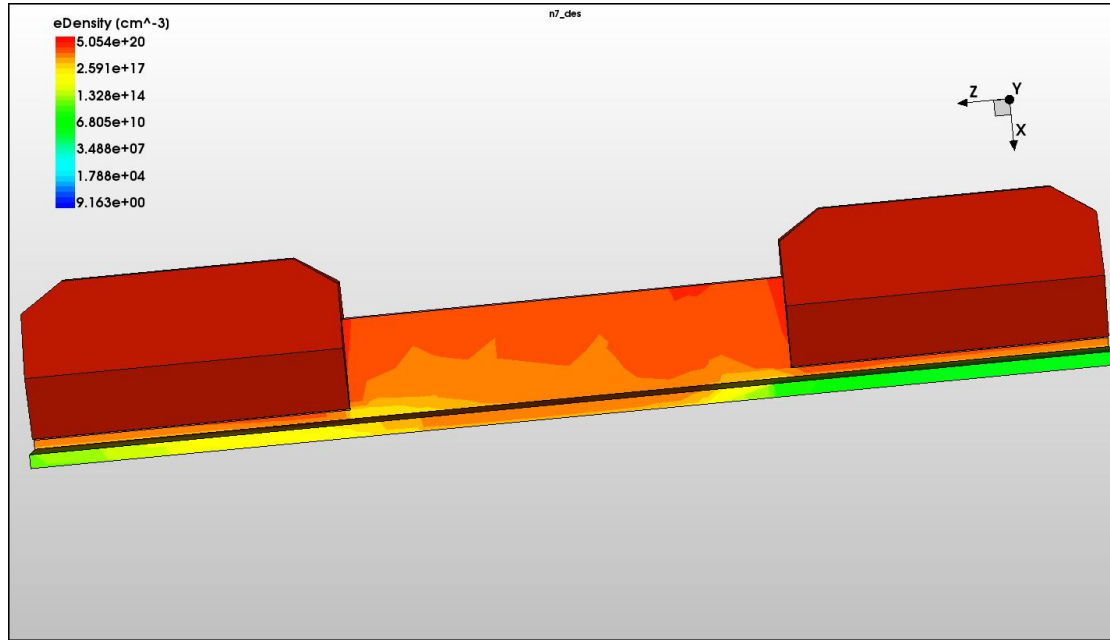
Channel Length = 25 nm.



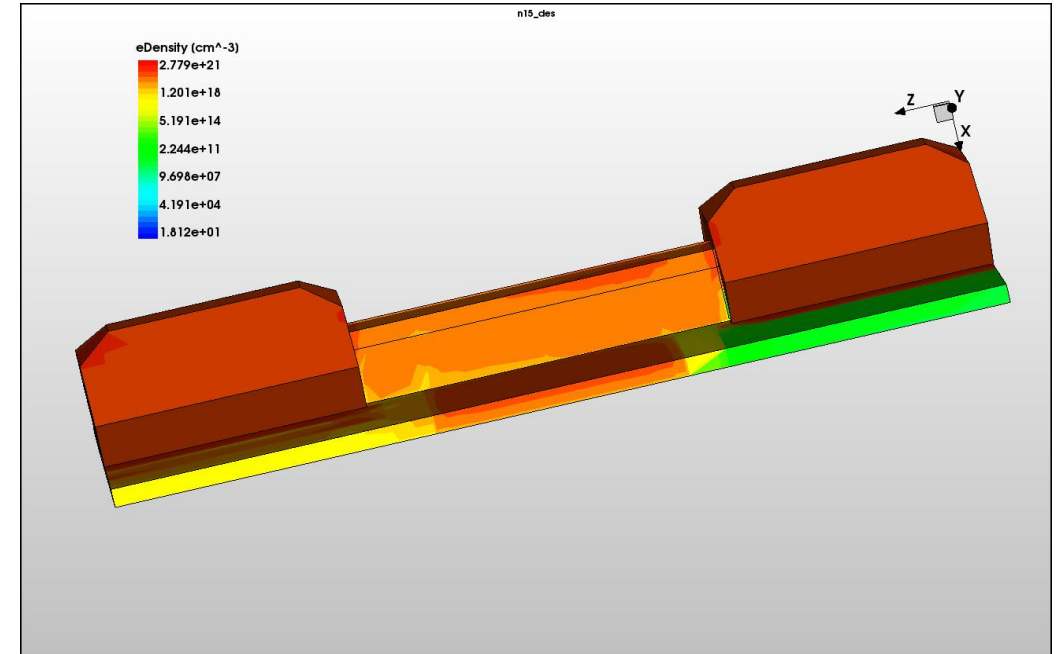


Electron Density Contour

Figure describes the electron density distribution across the channel and source and drain.



Electron density distribution of the conventional Si channel device.

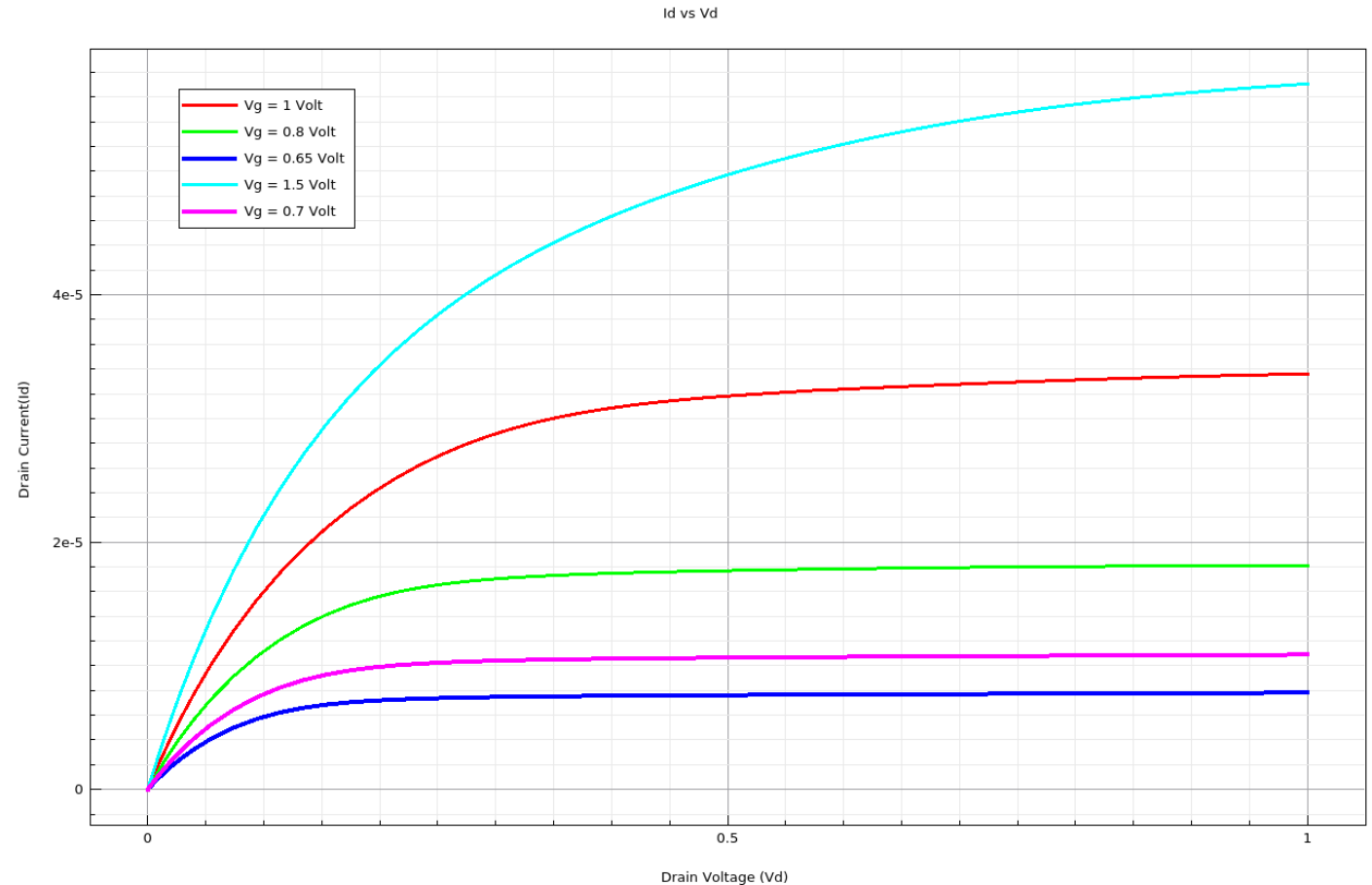


Electron density distribution of the heterojunction Si-SiGe channel device.

- For both the devices, source/drain is much highly doped than that of channel region.
- Out of both the devices, heterostructure device has high electron distribution.

Result Discussion:

- Plot represents behaviour of Drain Current(I_D) with change of Drain Voltage(V_D) at different Gate Voltages(V_g) for heterojunction Si-SiGe device.

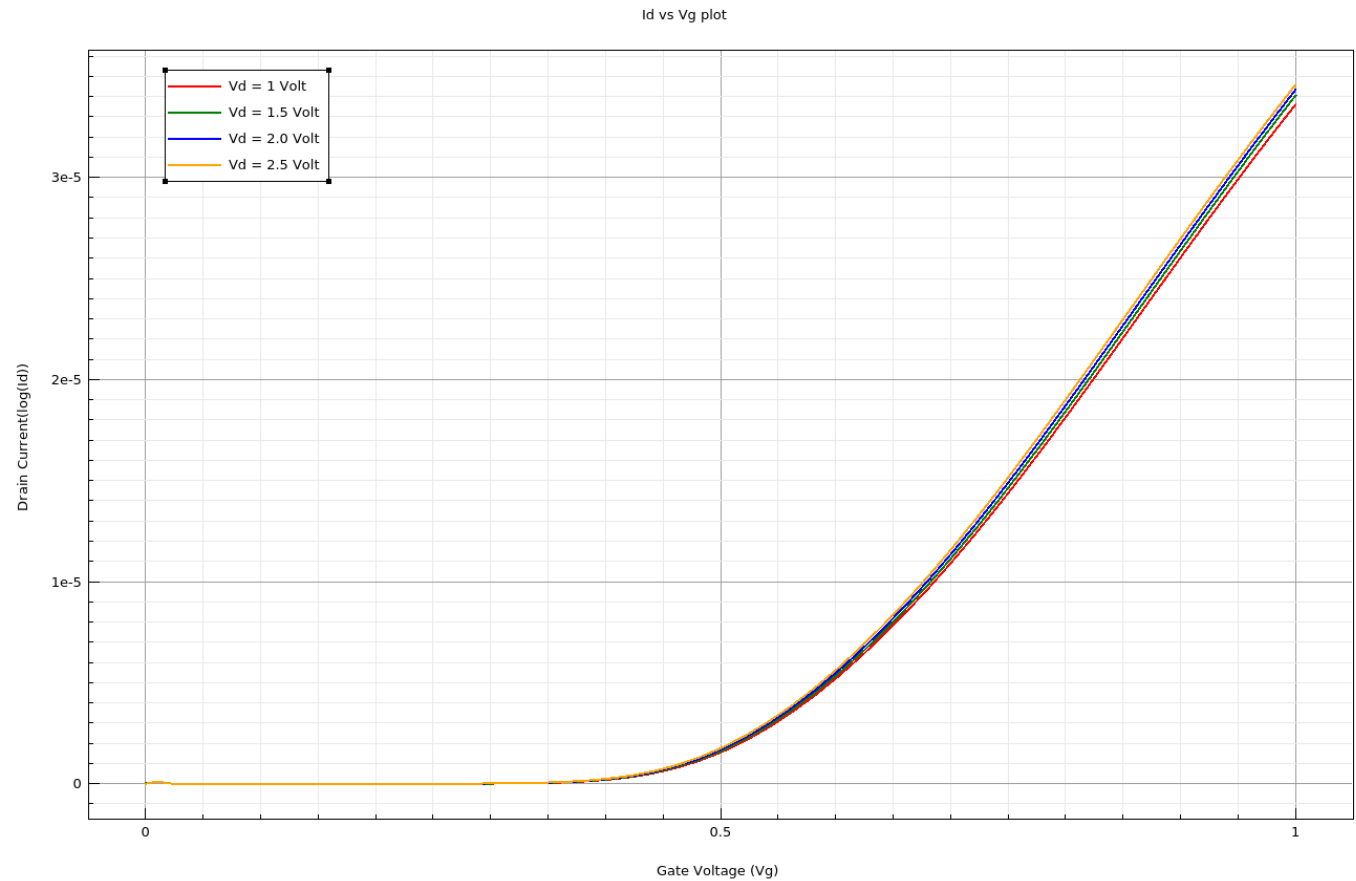


Contd..

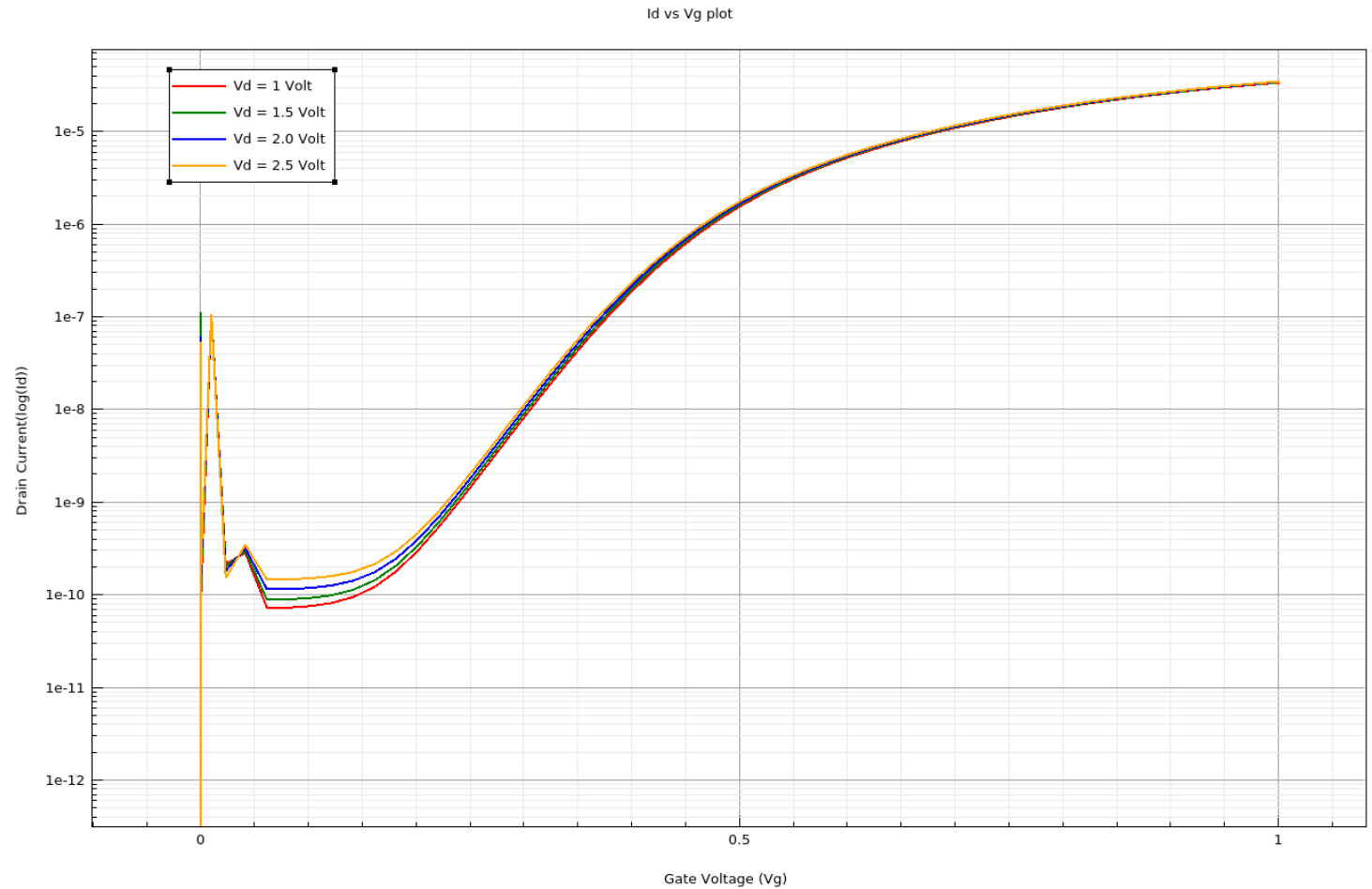
- Plot of Drain Current(I_d) vs Gate Voltage(V_g) at different Drain Voltages for heterojunction Si-SiGe device.

We can observe,

$$V_{TH} = 0.572$$

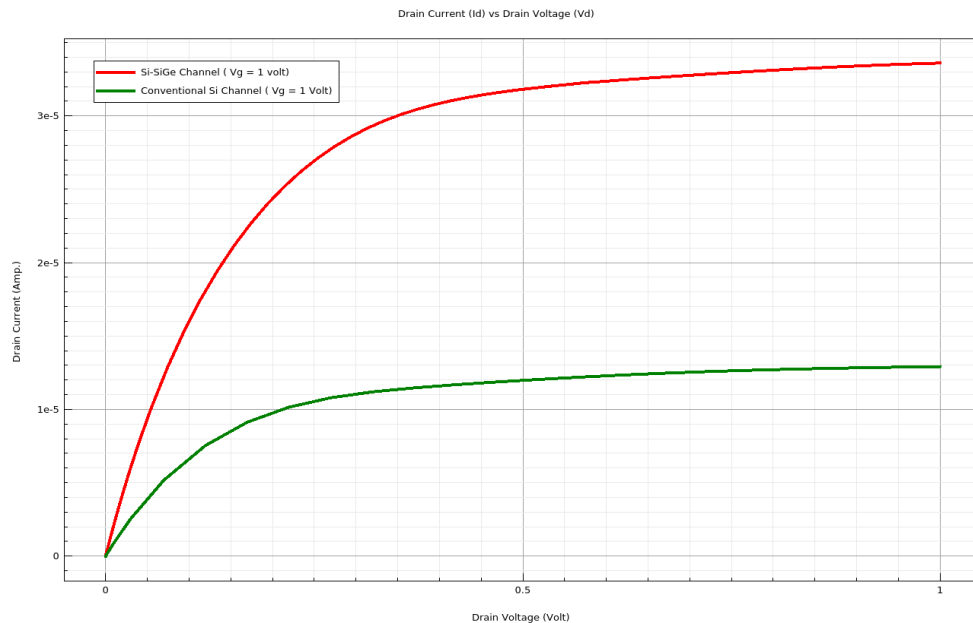


- Plot represents the behaviour of $\log(I_d)$ vs Gate Voltage (V_g) for different Drain voltages (V_d) for heterojunction Si-SiGe device.
- From the graph it can be observed that :
 - $\frac{I_{ON}}{I_{OFF}} = 10^5$



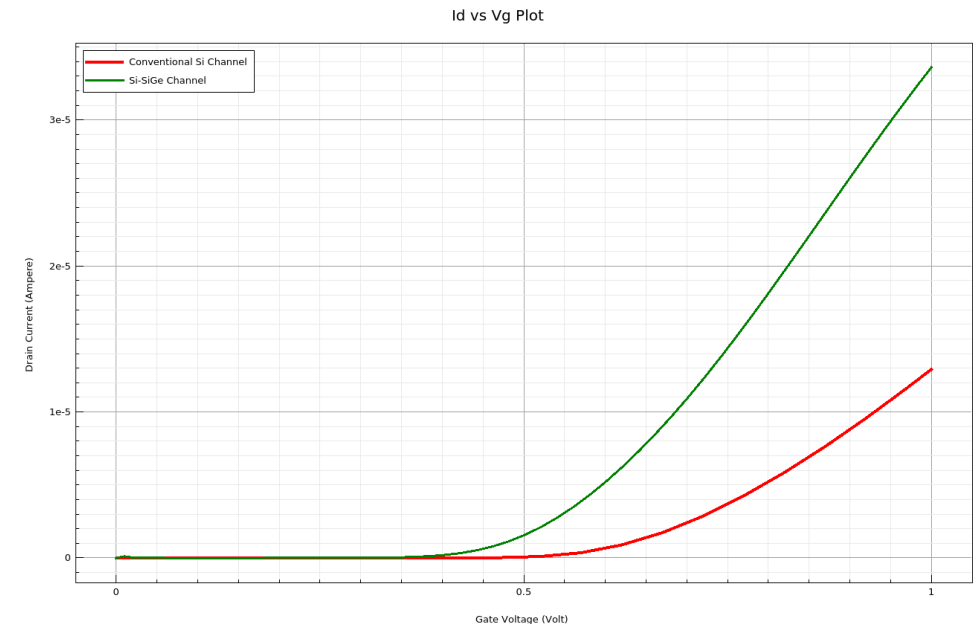
Comparative Analysis

■ OUTPUT CHARACTERISTICS OF BOTH DEVICES



- I_d vs V_d behaviour of both the structures at Gate Voltage (V_g) = 1 volt.

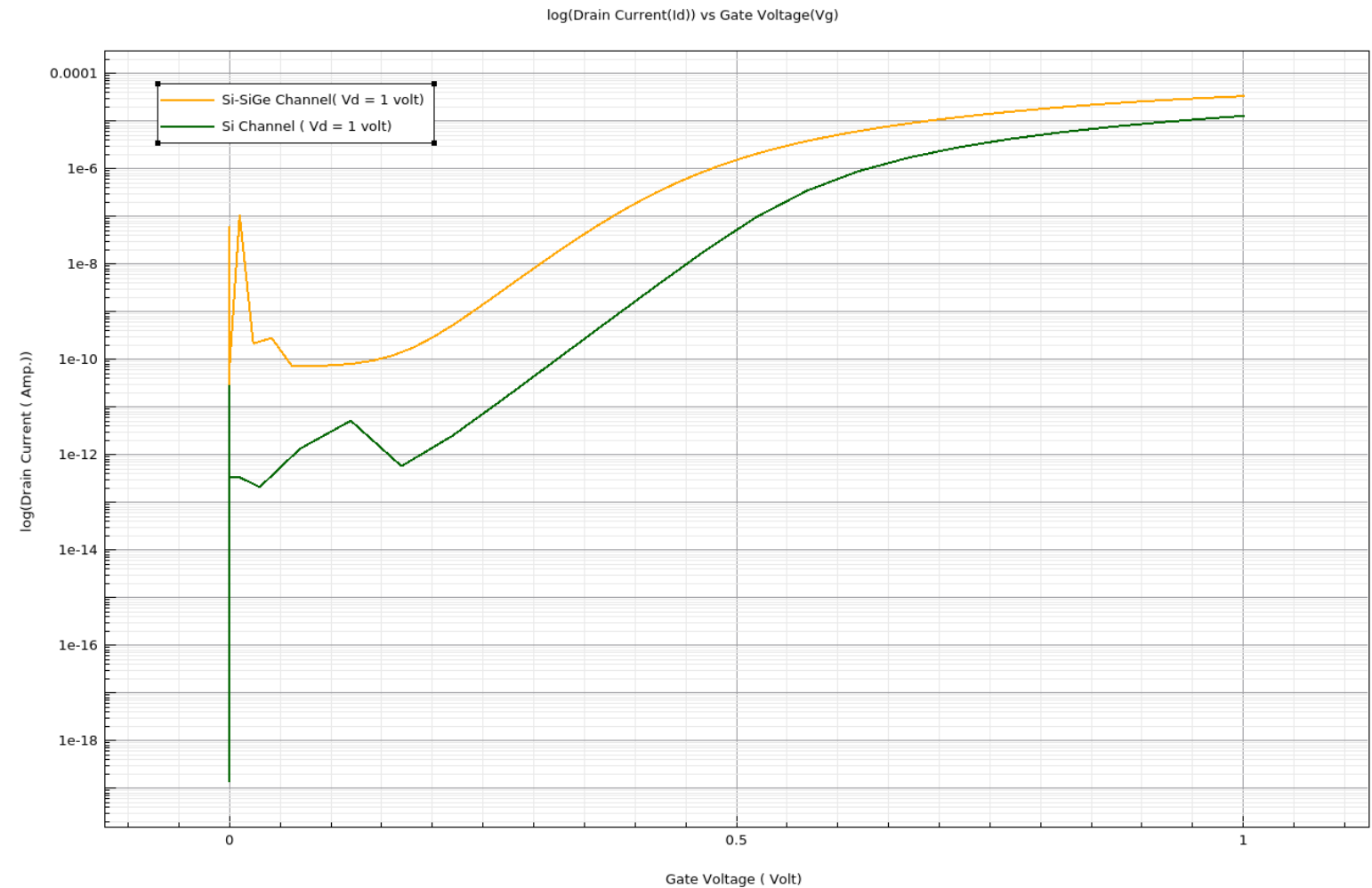
■ TRANSFER CHARACTERISTICS OF BOTH DEVICES



- I_d vs V_g behaviour of both the structures at Drain Voltage (V_d) = 1 volt.

Contd...

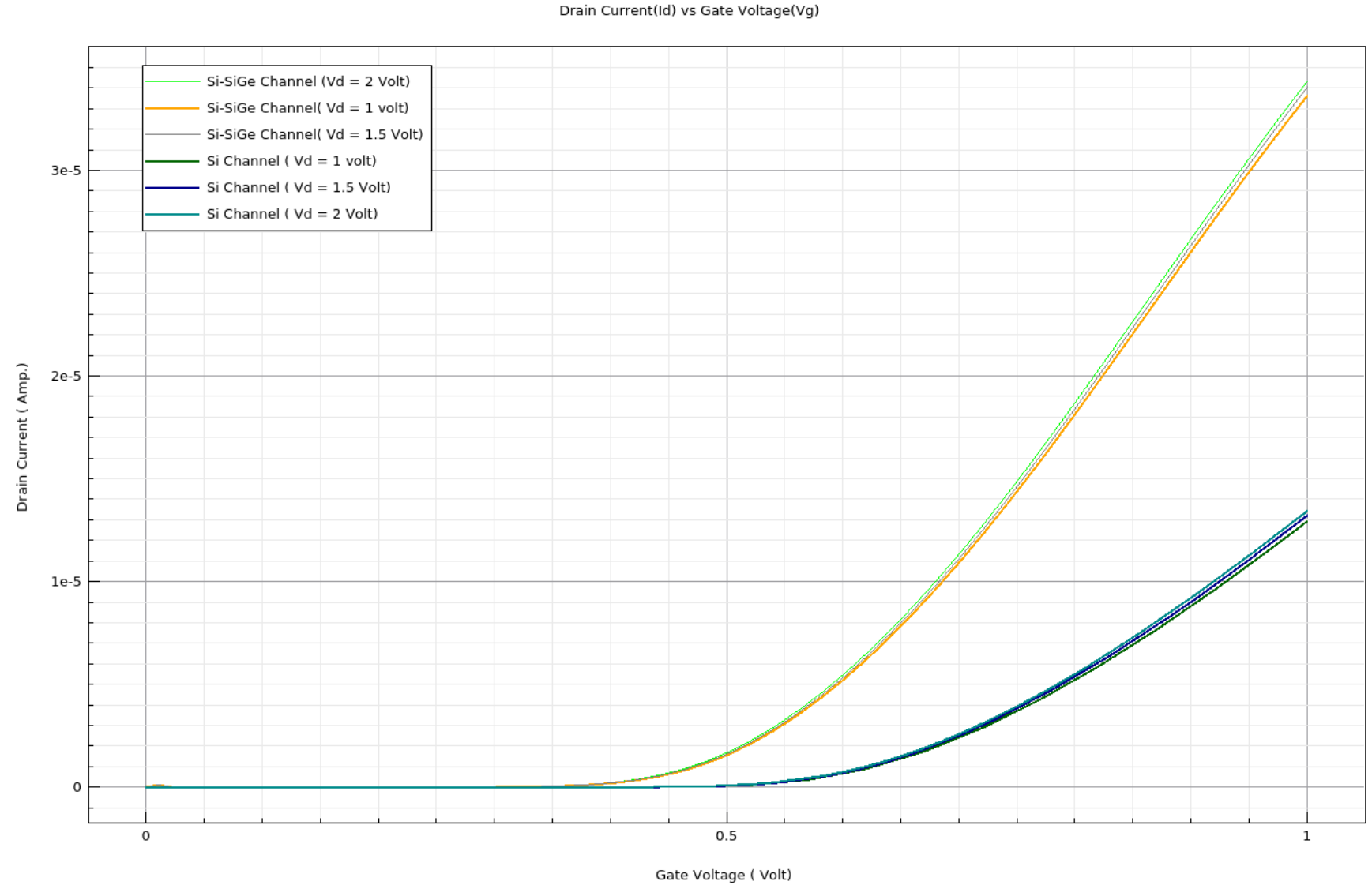
- Figure represents the behaviour of $\log(\text{Drain current}) (I_d)$ vs Gate Voltage (V_g) at constant Drain Voltage (V_d) = 1 volt for both devices.



Contd..

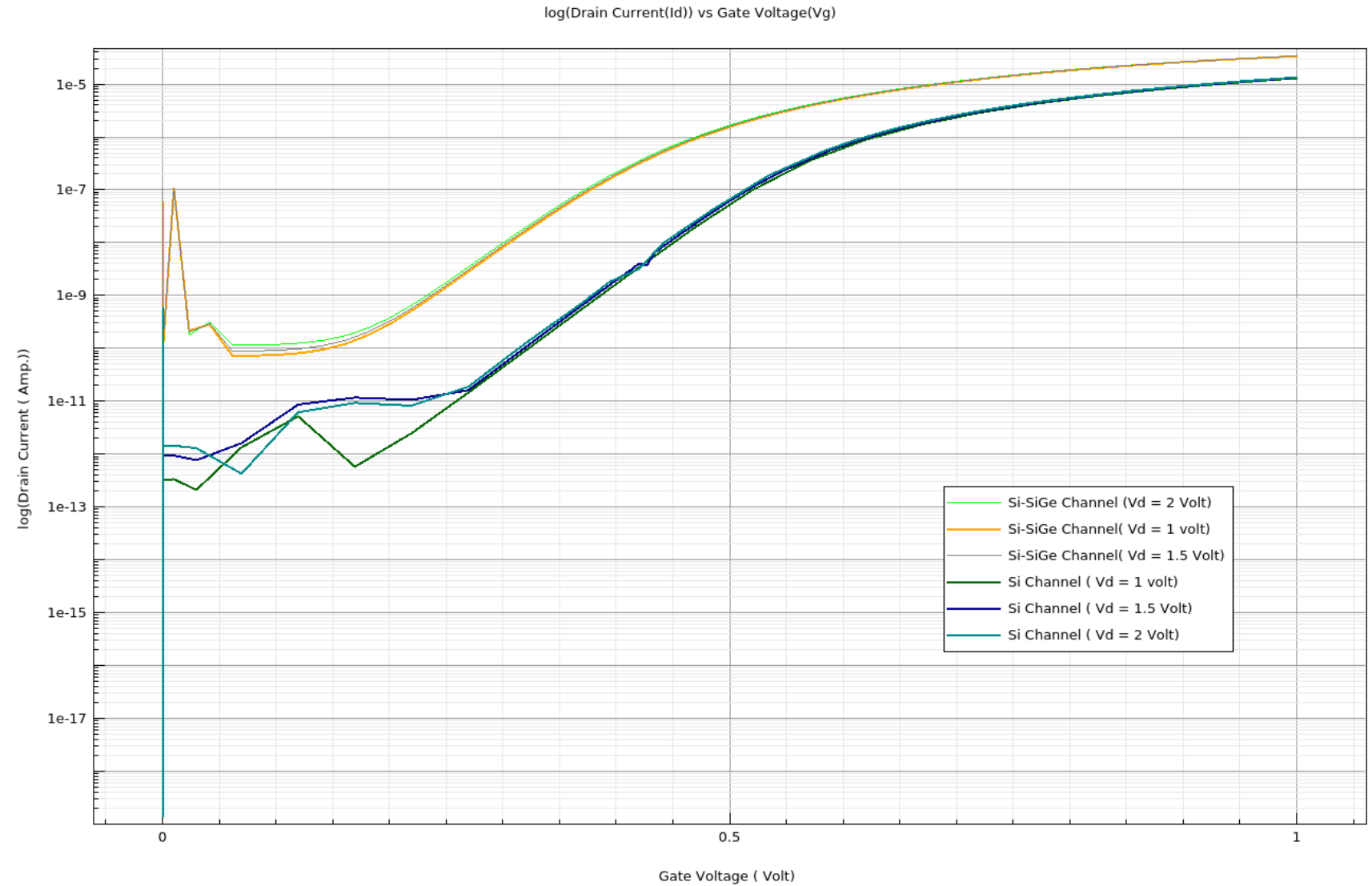
- Transfer characteristics of both the devices at different gate voltages.

Here, it can be observed that heterojunction Si-SiGe device has less Threshold Voltage.



Contd....

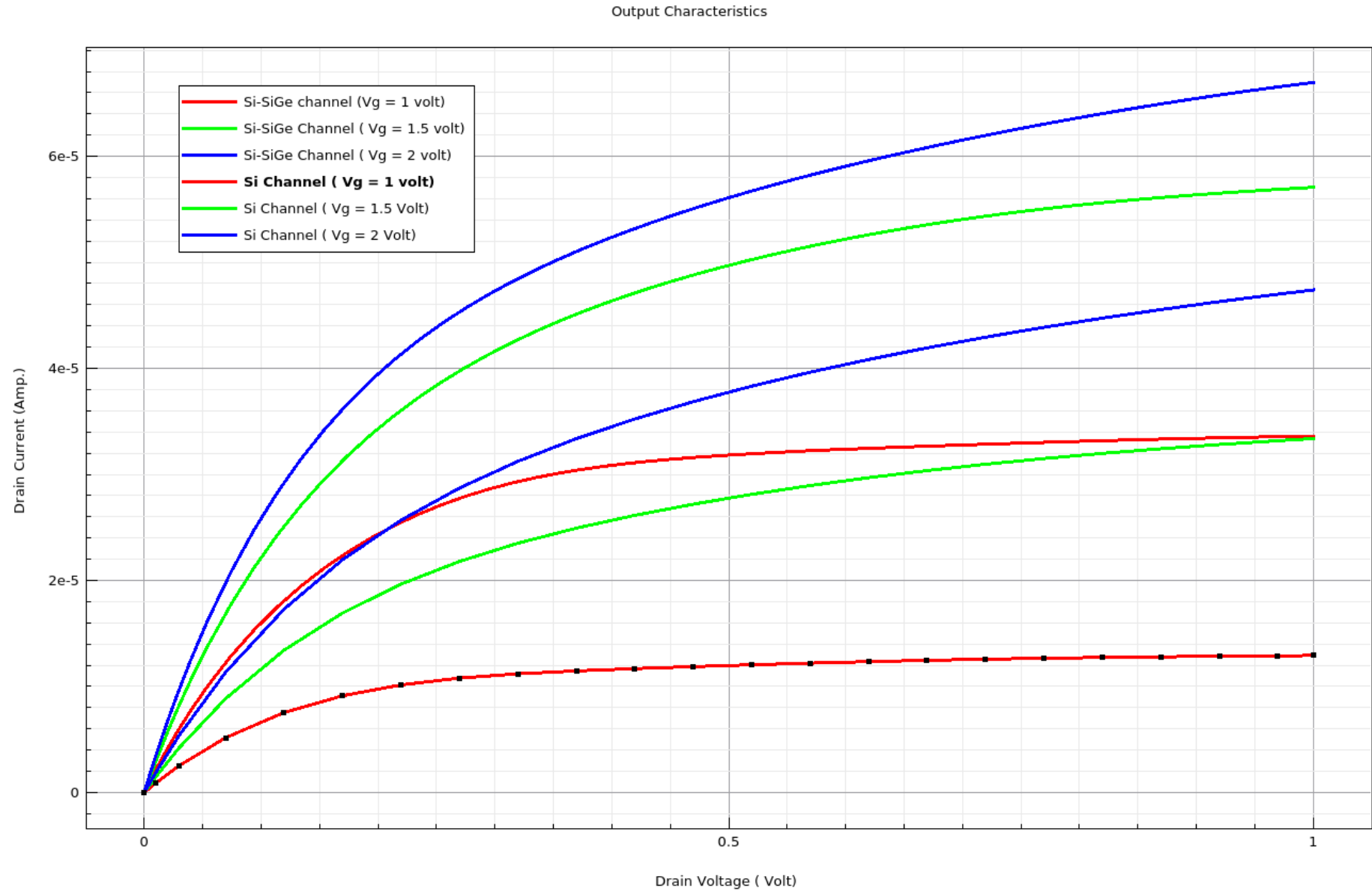
- Plot represents drain current(I_d) vs Gate Voltage(V_g) for different Drain Voltages (V_d) for both conventional Si and heterojunction Si-SiGe channel device.



Contd..

- Plot represents output characteristics of both devices for different Gate Voltages.

It can be observed, that for same Gate Voltage, heterojunction structure offers more drain current than that of conventional single channel device.



Calculation of DIBL:

DIBL can be calculated as :

$$DIBL = \frac{V_{th}(V_{DS} = V_{DS1}) - V_{th}(V_{DS} = V_{DS2})}{V_{DS2} - V_{DS1}}$$

where , $V_{DS2} > V_{DS1}$

For, conventional Si channel device

$$DIBL = (0.675 - 0.670) / (1.5 - 1.0)$$

So, DIBL = 10 mV/V

For heterojunction Si-SiGe channel device

$$DIBL = (0.573 - 0.570) / (1.5 - 1.0)$$

So, DIBL = 6 mV/V

Conclusion :

From different transfer and output characteristics of we can observe for conventional Si channel device :

- Threshold voltage (V_{th}) = 0.671 Volt
- Subthreshold Swing = 63 mV/Decade
- DIBL = 10 mV/V

From different transfer and output characteristics of we can observe for heterojunction Si-SiGe channel device :

- Threshold voltage(V_{th}) = 0.572 Volt
- Subthreshold Swing = 66 mV/Decade
- DIBL = 6 mV/V

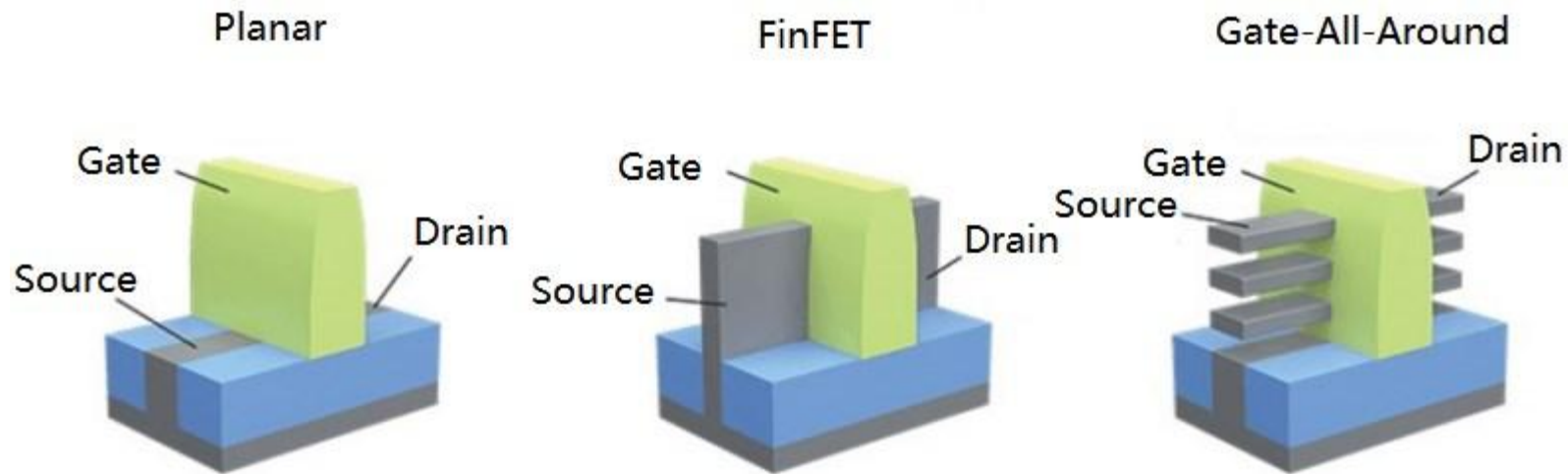
Conclusion :

- We can observe that using Si-SiGe structure there is about **17.3%** improvement in the Threshold Voltage.
- Improvement in DIBL of **~40%** in heterojunction device can also be observed.
- Also, we can observe significant increase in the on-current for the heterostructure Si/SiGe device .

Future Work

- Virtual Fabrication of Si/Si-Ge hetero structure based FinFET for different mole fraction, different channel height and stress analysis.

Additionally, structures like GAA(Gate-All-Around)FET, CFET can be explored.



References :

1. D.Guo, G.Karve, K-Ylim, “FINFET technology featuring high mobility SiGe channel for 10nm and beyond”, 2016 IEEE Symposium on VLSI Technology.
2. Matteo Pelosi, “From FinFET to Nanosheet Si-SiGe GAAFET”.
3. Archana, Manish Verma, Dhandapani Vaithiyanathan,“ Investigation of FinFET with multi-channel structures on device characteristics”, 2024 IEEE International Conference of Electron Devices Society Kolkata Chapter (EDKCON).
4. Jean-Pierre Colinge, “Fin-FETs and Other Multi-Gate Transistors”.

THANK YOU!